

Assessing human exposure risk to cadmium through inhalation and seafood consumption.

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The role of cadmium (Cd) bioaccessibility in risk assessment is less well studied. The aim of this study was to assess human health risk to Cd through inhalation and seafood consumption by incorporating bioaccessibility. The relationships between trophically available Cd and bioaccessibility were constructed based on available experimental data. We estimated Cd concentrations in human urine and blood via daily intake from seafood consumption and inhalation based on a physiologically-based pharmacokinetic (PBPK) model. A Hill-based dose-response model was used to assess human renal dysfunction and peripheral arterial disease risks for long-term Cd exposure. Here we showed that fish had higher bioaccessibility (~83.7%) than that of shellfish (~73.2%) for human ingestion. Our results indicated that glomerular and tubular damage among different genders and smokers ranged from 18.03 to 18.18%. Our analysis showed that nonsmokers had 50% probability of peripheral arterial disease level exceeding from 3.28 to 8.80%. Smoking populations had 2-3 folds higher morbidity risk of peripheral arterial disease than those of nonsmokers. Our study concluded that the adverse effects of Cd exposure are exacerbated when high seafood consumption coincides with cigarette smoking. Our work provides a framework that could more accurately address risk dose dependency of Cd hazard. Copyright © 2012 Elsevier B.V. All rights reserved.